

Experiment No. 10

Version Control using Git & GitHub

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Roll No:	68		

Aim

To understand and implement Version Control using Git and GitHub by creating repositories, committing changes, and pushing code to a remote repository.

Theory

Git is a distributed version control system that tracks changes in source code during software development. It allows multiple developers to work together on non-linear development.

GitHub is a cloud-based hosting service that lets you manage Git repositories. It provides a web interface for collaboration, code review, and project management.

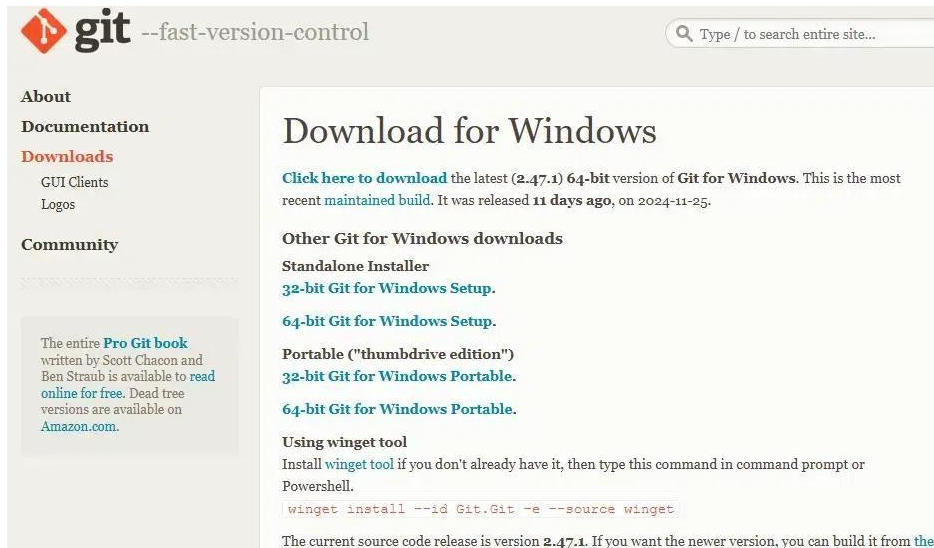
Key concepts include: Working Directory, Staging Area (Index), Local Repository, and Remote Repository. Changes flow from working directory → staging area → local repository → remote repository.

Step by Step Procedure

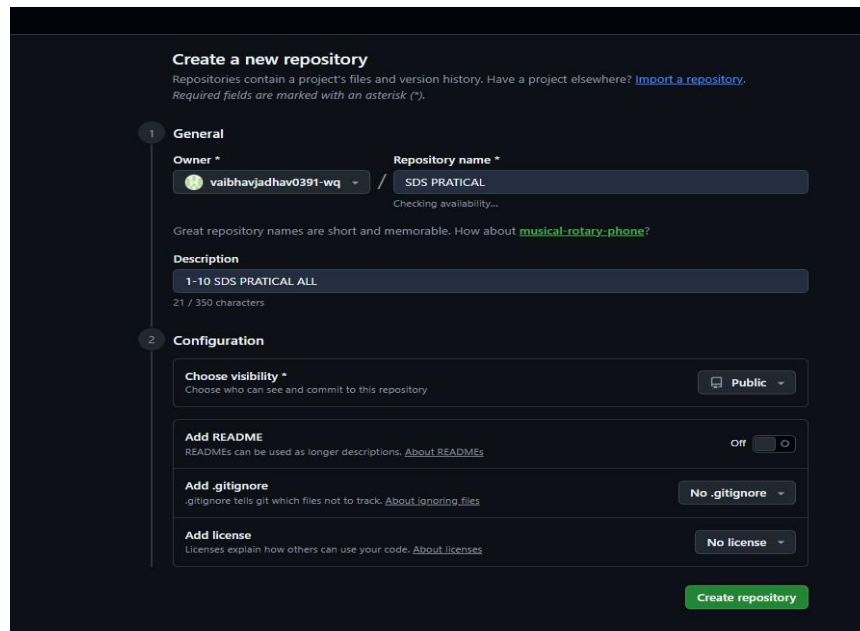
Sr.	Step Description
1.	Installation of Git
2.	Configure username & email
3.	Create local repository
4.	Create files
5.	Add files to staging
6.	Commit changes
7.	Create GitHub repository
8.	Link local repo to GitHub

9. Push files

Screenshots



1. installation verification (git version)



2. Repository creation

```
MINGW64/e/sds/vaibhav
vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds (master)
$ ls /e
123/
6TH/
A/
AJP-classTest2.docx
AVANTIKA JADHAV/
Alarms/
Android/
Animated ppt 2.pptx
AudioBooks/
Border-2.2026.1080p.V2-HDTC.Hindi-Line.x264-HDHub4u.Ms (2).mkv
CSS MICROPROJECT/
DCIM/
DIPLOMA /
Dhurandhar-The.Revenge.2026.1080p.HEVC.V2-HDTC.Hindi-Lin.mkv
Final Year Project Dairy for 6th Semester.rtf
Final Year Project report first pages.pdf
Java program /
LOST.DIR/
MAD Project/
Microproject Report Format (4).docx
Movies/
Mp.pptx
Music/
Notifications/
Objective.docx
Pictures/
Podcasts/
Presentation2.pptx
Recordings/
Ringtones/
STE/
System Volume Information/
This PC - Shortcut.lnk*
Time.docx
VirtualClassroom-master/
VirtualClassroom-master.zip
WirelessLan_ROG_Realtek_B_V6001.15.160.100SubI_44703.exe*
Word.lnk*
aikr/
aikr.zip
cpe.pdf
cpe.rtf
cpp/
cpr mit.docx
'doc submit to mit'/
honor/
'micro project final'/
miro/
mit/
ppt/
ppt no 283.pptx
project/
sds/
'shivani py.docx'
'ste 2/'
vaibhav 5th.pdf
yb.net/
'-$Animated ppt 2.pptx'
'-$Animated ppt.pptx'
'-$Mp.pptx'
'-$Presentation2.pptx'
'-$Wood Type.pptx'
'wRL0003.tmp'
'wRL1068.tmp'
'$$\360\237\230\201'' VAIBHAV.DOC'/
'$$\360\237\247\221\342\200\215\360\237\222\273''g'/

vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds (master)
$ cd /e
bash: cd /e: No such file or directory

vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds (master)
$ cd /e
ls
123/
6TH/
A/
AJP-classTest2.docx
AVANTIKA JADHAV/
Alarms/
Android/
Animated ppt 2.pptx
AudioBooks/
Border-2.2026.1080p.V2-HDTC.Hindi-Line.x264-HDHub4u.Ms (2).mkv
CSS MICROPROJECT/
DCIM/
DIPLOMA /
Dhurandhar-The.Revenge.2026.1080p.HEVC.V2-HDTC.Hindi-Lin.mkv
Final Year Project Dairy for 6th Semester.rtf
Final Year Project report first pages.pdf
Java program /
LOST.DIR/
MAD Project/
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Movies/
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Podcasts/
Presentation2.pptx
Recordings/
Ringtones/
STE/
System Volume Information/
This PC - Shortcut.lnk*
Time.docx
VirtualClassroom-master/
VirtualClassroom-master.zip
WirelessLan_ROG_Realtek_B_V6001.15.160.100SubI_44703.exe*
Word.lnk*
aikr/
aikr.zip
cpe.pdf
cpe.rtf
cpp/
cpr mit.docx
'doc submit to mit'/
honor/
'micro project final'/
miro/
mit/
ppt/
ppt no 283.pptx
project/
sds/
'shivani py.docx'
'ste 2/'
vaibhav 5th.pdf
yb.net/
'-$Animated ppt 2.pptx'
'-$Animated ppt.pptx'
'-$Mp.pptx'
'-$Presentation2.pptx'
'-$Wood Type.pptx'
'wRL0003.tmp'
'wRL1068.tmp'
'$$\360\237\230\201'' VAIBHAV.DOC'/
'$$\360\237\247\221\342\200\215\360\237\222\273''g'/

vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e
$ cd sds
```

```
MINGW64/e/sds/vaibhav
vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e
$ cd sds
vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds (master)
$ cd vaibhav
vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds/vaibhav (master)
$ git init
Initialized empty Git repository in E:/sds/vaibhav/.git/
vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds/vaibhav
$ git add .
fatal: detected dubious ownership in repository at 'E:/sds/vaibhav'
'E:/sds/vaibhav' is on a file system that does not record ownership
To add an exception for this directory, call:

    git config --global --add safe.directory E:/sds/vaibhav
vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds/vaibhav
$ git commit -m "Added PDFs"
fatal: detected dubious ownership in repository at 'E:/sds/vaibhav'
'E:/sds/vaibhav' is on a file system that does not record ownership
To add an exception for this directory, call:

    git config --global --add safe.directory E:/sds/vaibhav
vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds/vaibhav
$ git remote add origin https://github.com/vaibhavjadhav0391-wq/SDS-PRATICAL-.git
fatal: detected dubious ownership in repository at 'E:/sds/vaibhav'
'E:/sds/vaibhav' is on a file system that does not record ownership
To add an exception for this directory, call:

    git config --global --add safe.directory E:/sds/vaibhav
vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds/vaibhav
$ git config --global --add safe.directory E:/sds/vaibhav
vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds/vaibhav (master)
$ git add .
vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds/vaibhav (master)
$ git commit -m "Added PDFs"
[master (root-commit) c16341b] Added PDFs
 9 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 B4-72-01-VAIBHAV JADHAV.pdf
 create mode 100644 B4-72-02-VAIBHAV JADHAV.pdf
 create mode 100644 B4-72-03-VAIBHAV JADHAV.pdf
 create mode 100644 B4-72-04-VAIBHAV JADHAV (2).pdf
 create mode 100644 B4-72-05-VAIBHAV JADHAV.pdf
 create mode 100644 B4-72-06-VAIBHAV JADHAV.pdf
 create mode 100644 B4-72-07-VAIBHAV JADHAV.pdf
 create mode 100644 B4-72-08-VAIBHAV JADHAV.pdf
 create mode 100644 report.pdf
vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds/vaibhav (master)
$ git remote add origin https://github.com/vaibhavjadhav0391-wq/SDS-PRATICAL-.git
vaibhav Jadhav@LAPTOP-ME3JP5TN MINGW64 /e/sds/vaibhav (master)
```

```

create mode 100644 84-72-08-VAIBHAV JADHAV.pdf
create mode 100644 report.pdf

vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (master)
$ git remote add origin https://github.com/vaibhavjadhav0391-wq/SDS-PRATICAL-.git
vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (master)
$ ^[[200~git branch -M main
bash: $'\E[200~git': command not found
vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (master)
$ git branch -M main
vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (main)
$ git push -u origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 16 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (11/11), 805.41 KiB | 14.13 MiB/s, done.
Total 11 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/vaibhavjadhav0391-wq/SDS-PRATICAL-.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (main)
$

```

3. Successful push

```

branch 'main' set up to track 'origin/main'.

vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (main)
$ git log
commit e563d1b8754b93348623ba45baccfad1411ea4b1d (HEAD -> main, origin/main)
Author: Vaibhav Jadhav <vaibhavjadhav0391@gmail.com>
Date:   Mon May 4 19:45:20 2026 +0530

    Added PDFs

vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (main)
$

```

4. Display commit logs (git log)

The screenshot shows a GitHub repository page for 'vaibhavjadhav0391-wq / SDS-PRATICAL-'. The repository is public and has 1 commit. The commit log shows a single commit by vaibhavjadhav0391-wq, titled 'Added PDFs', with a commit hash of e563d1b. The commit message is 'Added PDFs'. The repository contains 10 files, all named '84-72-01-VAIBHAV JADHAV.pdf' through '84-72-08-VAIBHAV JADHAV.pdf', and a 'report.pdf'. All files were added 4 minutes ago. The repository also has a 'README' section with a button to 'Add a README'.

5. Show files on GitHub repo

Questions

1. What is the difference between Git and GitHub?

Git and GitHub are related but different tools used in software development.

- Git is a distributed version control system. It is used to track changes in files and manage different versions of a project locally on a computer. It allows developers to work offline and maintain history of changes.
 - GitHub is a cloud-based platform that hosts Git repositories. It provides features like collaboration, sharing code, issue tracking, and version management over the internet.
-

2. Explain the difference between git add and git commit

- **git add:**
This command is used to move changes from the working directory to the staging area. It prepares the files for saving but does not actually save them.
 - **git commit:**
This command saves the staged changes permanently in the repository. Each commit creates a snapshot of the project with a message describing the changes.
-

3. What is the staging area in Git?

The staging area is an intermediate area in Git where changes are stored before they are committed.

- It allows the user to select specific changes to include in the next commit
 - It acts as a buffer between working directory and repository
 - It helps in organizing commits properly
-

4. What happens if you modify a file after committing it?

If a file is modified after committing:

- The new changes are not automatically saved in the repository
 - Git marks the file as “modified”
 - You must again use git add and git commit to save the new changes
-

5. Difference between git push and git pull

- **git push:**
It is used to upload local commits from your system to a remote repository like GitHub. It updates the online repository.
 - **git pull:**
It is used to fetch and download changes from the remote repository to your local system. It keeps your project updated.
- .

6. What is Version Control? Differentiate between Centralized and Distributed VCS

Version Control is a system that records changes made to files over time so that previous versions can be restored when needed.

Centralized Version Control System (CVCS):

- Has a single central server
- All users connect to this server
- Requires internet connection
- If server fails, data may be lost
- Example: SVN

Distributed Version Control System (DVCS):

- Every user has a complete copy of the repository
- Works offline
- Faster and more secure
- No single point of failure
- Example: Git

Conclusion

Git and GitHub are powerful tools used for managing and sharing projects. Git helps in tracking changes and maintaining version history, while GitHub allows collaboration and remote storage. Version control systems improve productivity, ensure data safety, and make teamwork easier in sof